1.0 Basic Algebra MA1905

1.1 Define the terms:

1.1.1 Constants
1.1.2 Variables
1.1.3 Equations
1.1.4 Functions

Constant: A quantity whose value does not change

For example, in the expression $4x + 12$, the number 12 is a constant.

Variable: A letter or symbol representing a value that can change. Common variables include x and y.

In the expression $4x + 12$, x is the variable.

For example, if $x = 5$, then $4(5) + 12 = 20 + 12 = 32$.

Equation: A mathematical statement showing that two expressions are equal. An equation contains an equal sign (=).

For example, $3x + 5 = 20$ is an equation. However, $3x + 5$ is an expression, not an equation.

Function: A relationship where each input has exactly one output. A function is often written using $f(x)$.

For example, if $f(x) = 2x + 3$, then $f(4) = ?$

Step 1: Replace x with 4.

$$f(\mathbf{4}) = 2(\mathbf{4}) + 3$$

Step 2: Calculate.

$$f(4) = 8 + 3 = 11$$

Therefore, $f(4) = 11$.

Think of a function like a simple process in a computer.

Input → Rule/Process → Output

Examples

Identify the variable and constants in $7x + 15 = 50$.

If $y = 8$, calculate $3y + 4$

For $f(x) = 3x - 2$, find $f(5)$.

Is $6x + 4$ an equation?

1.2 Express values using exponents and scientific notations

Exponents

An exponent tells us how many times a number or variable is multiplied by itself.

$$2^4 = 2 \times 2 \times 2 \times 2 = 16$$

In words, we would say "2 to the power (exponent) of 4".

The 2 is the base.

The 4 is the exponent.

Write in exponent form.

$2 \times 2 \times 2$ $4 \times 4 \times 4 \times 4 \times 4$ $(-5) \times (-5)$

$(-x)(-x)(-x)$ $3 \cdot x \cdot y \cdot y \cdot 3 \cdot x \cdot x \cdot x$ $a \cdot b \cdot b \cdot a \cdot b$

Zero Power Rule

Any number or expression raised to the zero power will always be 1.

$$x^0 = 1$$

$$10^0 = 1$$

Why is this? Let's list some powers of 3, starting at 3^4, and then taking one away from the exponent each time. We will stop at an exponent of 0.

Negative Exponent Rule

Let's review and continue our list from earlier starting at 3^4, and then taking one away from the exponent each time. This time, we will not stop at 0.

Generally, if we think of our expression as a fraction, negative exponents in the numerator must be moved to the denominator. Similarly, negative exponents in the denominator need to be moved to the numerator. When the base with the exponent moves, the exponent is now positive.

$$x^{-1} = \frac{1}{x}$$

$$10^{-1} = \frac{1}{10} = 0.1$$

Laws of Exponents

The Product Rule

When we multiply exponential expressions with the same base, we simplify by adding the exponents.

$$x^a \cdot x^b = x^{a+b}$$

Evaluate.

$3^4 \times 3^7$

$5^8 \cdot 5^{-6}$

Simplify.

$x^2 \cdot x^4$

The Quotient Rule

When we divide exponential expressions with the same base, we simplify by subtracting the exponents ($x \neq 0$ since division by zero is undefined).

If the larger exponent is in the numerator: $\frac{x^a}{x^b} = x^{a-b}$

If the larger exponent is in the denominator: $\frac{x^a}{x^b} = \frac{1}{x^{b-a}}$

Evaluate.

$\frac{2^7}{2^5}$

$\frac{3^4}{3^5}$

$\frac{x^6}{x^6}$

The Power Rule

To raise a power to a power, keep the same base and multiply the exponents.

$$(\boldsymbol{x^a})^{\boldsymbol{b}} = \boldsymbol{x^{ab}}$$

Evaluate.

$(2^2)^3$

$(x^{-3})^5$

Scientific Notation

Scientific notation is a convenient way to write very large or very small numbers in decimal form using powers of 10.

To write numbers in scientific notation:

- Move the decimal point so there is one nonzero digit to the left of the decimal point.
- Multiply the result by a power of ten with an exponent equal to the number of places the decimal point was moved. The exponent is positive if the decimal point has been moved to the left. The exponent is negative if the decimal point has been moved to the right.

A number is written in scientific notation if it is in the form $A \times 10^n$, where $1 \leq A < 10$ and n is an integer.

For example,

$$310\ 000\ 000 = 3.1 \times 10^8$$

$$0.0000042 = 4.2 \times 10^{-6}$$

Convert the numbers from standard form to scientific notation.

4 567 000 0.000461

157 000 000

Convert the numbers from scientific notation to standard form.

2.45×10^{-2} 3.1426×10^3

1.3 State the order of operations for mathematics calculations

Consider the expression $3 + 4 \cdot 2$

a) If the addition is performed first, the answer is _____.

b) If the multiplication is performed first, the answer is _____.

Both answers cannot, of course, be correct.

In this problem, the correct answer is_____.

When dealing with arithmetic or algebraic expressions, the following convention is followed.

- Brackets are evaluated first. If more than one set is present, we evaluate from the inside out.
- Exponents
- Multiplication and Division (from left to right)
- Addition and Subtraction (from left to right)

Note

This sequence of calculations is often referred to as BEDMAS.

Evaluate each expression below. Show all calculations.

$a)\ 12\ \times 5 - 2$

$b)\ 12 - 5 \times 2$

$c)\ 10 + 16 \div 2^3$

$d)\ 10 + (8 \div 2)^4$

$e)\ 6 - 2(3 + 2^2)$

$f)\ 8 + 2(3 + 2)^2$

$g)\ 4[5 - 2(7 - 3)] + 1$

$h)\ 3 + 2(5 - 3^2)^2 - 2$

$i)\ \dfrac{4 \times 72 + 3^3}{(200 - 5^3) + 5 \times 6}$

1.4 Solve problems requiring the calculation of percentages

Percent means “for every one hundred”. This means a percent can be expressed as a fraction with a denominator of 100.

32% means 32 out of 100 or $\frac{32}{100}$.

Convert among decimal and percent forms

To convert a decimal to a percent:

- Multiply the decimal value by 100
- Write the percent symbol (%) after the number.

Express the following as a percent.

0.02

120

0.000552

To convert a percent to a decimal:

- Divide the percent by 100
- Remove the percent symbol

Express the following as a decimal.

106%

1.06%

0.023%

Calculate the percentage one number is of another number

- Write a fraction using the two numbers. The number after the word “of” is always the denominator and the other number is the numerator
- Use your calculator to convert the fraction to a decimal
- Express the decimal as a percent

Essentially, we are given the part and the whole. To find the percentage,

$$Percentage = \frac{part}{whole}$$

For example, 11 is what percent of 50?

$$Percentage = \frac{part}{whole}$$

$$Percentage = \frac{11}{50}$$

$$Percentage = 0.22\ (\times 100)$$

$$Percentage = 22\%$$

Therefore, 11 is 22% of 50.

What percent of 75 is 12?

42 is what percent of 25?

Calculate the percent of a number

We are given the percentage and the whole. To find the part,

$$Part = percentage \times whole$$

For example, what is 20% of 35?

$$Part = percentage \times whole$$

$$Part = 20\% \times 35$$

$$Part = 0.20 \times 35$$

$$Part = 7$$

Therefore, 7 is 20% of 35.

What is 15% of $120?

What is 29% of 106.55?

Calculate a number when a percentage of it is known

We are given the part and the percent. To find the whole,

$$Whole = \frac{part}{percentage}$$

For example, 30 is 60% of what number?

$$Whole = \frac{30}{60\%}$$

$$Whole = \frac{30}{0.60}$$

$$Whole = 50$$

Therefore, 30 is 60% of 50.

25 is 33.33% of what number?

230 is 115% of what number?

Solve. Show all steps. Clearly state your final answer.

Suppose you answered 37 out of 40 multiple choice questions correctly on a quiz. What grade did you receive on the quiz? Round your answer to the nearest percent.

Note: Assume each question is worth 1.

Suppose you buy a book for $9.99. The HST (Harmonized Sales Tax) is 13%. What is the total cost of the book, including tax?

A recent survey showed that 35% of Canadian college campuses are in rural or remote communities. If 476 campuses are in rural or remote communities, what is the total number of Canadian college campuses?

Extra Practice

1. Convert from a percent to a decimal.

a. 3% b.17% c. 200%

2. Convert from a decimal to a percent.

a. 0.009 b. 0.57 c. 1.25

3. Solve.

a. What is 50% of 120? b. What is 120% of 80?

c. What % is 2 out of 20? d. What % is 125 out of 50?

e. 33 is 11% of what number? f. 2 is 8% of what number?

Solve. Show all steps. Clearly state your final answer**.**

a. A breakfast cereal provides 55% of the daily requirement of iron. What percentage of iron must be found in other foods?

b. A customer's purchase, before tax, comes to $162.54. What is the total including 15% sales tax?

c. In a closing-out sale, a store reduces its prices by 70%. What is the sale price for a radio that originally cost $40?

d. Emma's car insurance is reduced by 5% for each year that she has not had an accident. The normal cost of her insurance is $620, but she has a 3-year "no claims bonus". How much is her insurance?

e. A new TV costs $375. You pay a 35% deposit. How much is left to pay?

1.5 Solve problems on a calculator using mathematical calculations for:

1.5.1 Real numbers
1.5.2 Exponents
1.5.3 Percentages

Real Numbers

Real numbers ($\mathbb{R}$) are all the numbers that can be placed on a number line. They include both rational and irrational numbers.

Number System	Symbol	Meaning	Examples
Natural numbers	$\mathbb{N}$	Counting numbers	1, 2, 3, 4, …
Whole numbers		Natural numbers plus zero	0, 1, 2, 3, ...
Integers	$\mathbb{Z}$	Whole numbers and their negatives	…-2, -1, 0, 1, 2…
Rational numbers	$\mathbb{Q}$	Numbers that can be written as a fraction of two integers	$\frac{3}{4}$, -2, 0.6, 1.25
Irrational numbers		Numbers that cannot be written as a fraction of two integers; their decimals do not terminate or repeat	$\sqrt{2}, \sqrt{5}, \pi$

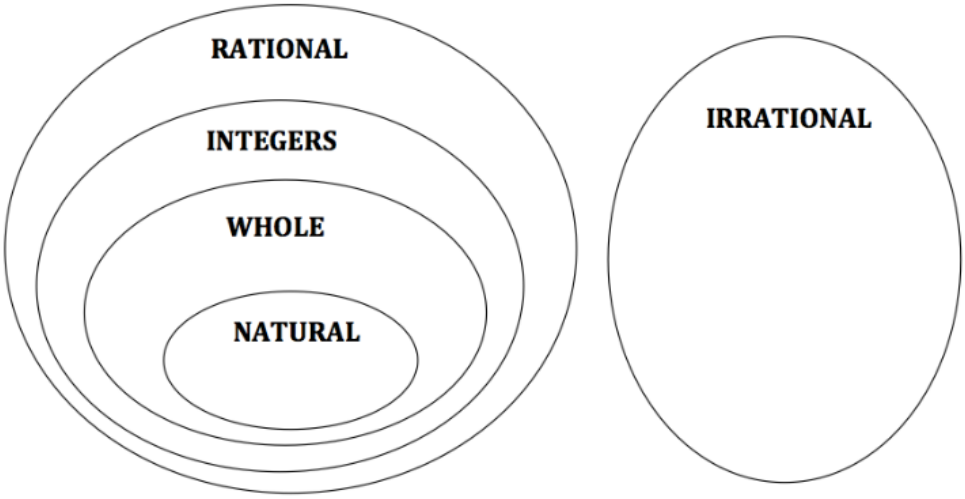
The number systems overlap.

For example,

5 is a natural number, whole number, integer, rational number, and real number.

-3 is an integer, rational number, and real number.

Pi is irrational and real.



To which number systems do the following belong? Check all that apply.

Number	Natural	Whole	Integer	Rational	Irrational	Real
3.19						
$-\frac{3}{8}$						
0						
-10						
0.272727…						
$\sqrt{17}$						

Real Number Calculations

Evaluate. Use the order of operations.

a) -8.4 + 15.7

b) 7.25 × 4.6

c) 20 - 4.5 × 3

d) (20 - 4.5) × 3

e) -12.6 + 8.9

f) 45.6 ÷ 3.8

g) 18 - 2.5 × 4

h) (18 - 2.5) × 4.

Exponents

The Power Rule (extension)

To raise a product to a power, each factor inside the parentheses must be raised to the power.

$$(xy)^a = x^a y^a$$

To raise a quotient to a power, the numerator and denominator inside the parentheses must be raised to the power.

$$\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$$

Examples

a) $(5x^2)^3$

b) $(2x^3)^4$

c) $\left(\frac{2x}{3}\right)^2$

d) $(3x)^2\ (2x^3)^4$

e) $(3x^{-1})\ (12x)^{-1}$

f) $\left(\frac{4}{x}\right)^2\ \left(\frac{2}{x}\right)^{-3}$

g) $(a^{-6}b)^{-1}(a^3b^{-1})^{-2}$

Review of Algebraic Expressions and Equations

A variable represents an unknown quantity.

A term is a number, variable or a product of numbers and variables.

9 $6ab$ y

Algebraic expressions are made up of terms.

$8x$ is an algebraic expression with one term

$x + 5$ has two terms: x and +5

$5a - 3b$ also has two terms: $5a$ and $-3b$

Terms with the **same variables raised to the same exponents** are called like terms. Otherwise, they are unlike terms.

Like Terms	Unlike Terms	
$3x, 2x$	$5x, 5x^2$	Same variable x, but different exponents
$-6x^2y, 2x^2y, 4x^2y$	$7y, 3z, 8x$	Different variables
$2ab^2c^3, ac^3b^2$	$6abc^3, 6ab^2$	Different variables and different exponents

Determine whether the pairs of terms are like terms.

a) $2x^2y$ and $3x^2y$

b) $4x^2y^2$ and $7x^2y^3$

c) $3xyz^4$ and $3x^4yz$

d) $17x^3y^4z^5$ and $8x^3y^4z^5$

Only like terms of algebraic expressions can be combined when adding or subtracting. When given an expression, you may rearrange the terms so that like terms are next to one another.

Simplify.

a) $3x + 2x$

b) $5x^2 - x^2$

c) $2x^2 + 5x - 3x^2 - x$

d) $4ab + 3bc - 5ab + 5b^2 - 2bc$

e) $2xy + 4x^2y - 6xy^2 + 8x^2y$

To multiply an algebraic expression by a real number,

- Using the distributive property, multiply the coefficients but leave the variables unchanged.
- Simplify by combining like terms.

Multiply and simplify.

a) $6(a + b)$

b) $-3(2x - y - 5)$

c) $-(a + 4b - 2c)$

d) $2(x + y) - 5(x - y)$

Add and subtract polynomials

To add polynomials, combine any like terms.

To subtract polynomials, we must be careful with the minus sign in front of a set of parentheses. Think of the minus sign as being negative. We must distribute it which will change the signs of all terms inside the parentheses. Then, combine any like terms.

Note:

Most students do not make mistakes with the variables or their exponents. The problem tends to be with the coefficients and their signs.

Simplify.

a) $(7x^2 - 6x + 3) - (5x^2 - 8x - 12)$

b) $(2x^2 - 4x + 3) + (5x^2 - 6x + 1)$

c) $(-6x^2y - 3xy + 7xy^2) - (5x^2y - 8xy - 5x^2y^2)$

Equations

An **equation** uses an equals sign (=) and indicates that two expressions are equal. An **algebraic equation** is an equation that contains at least one variable.

A linear equation is one that takes the form $ax + b = c$ *(where a, b, and c are constants)*

For example, $x + 1 = 9$ and $x - 5 = 2$ are linear equations.

The Addition (Subtraction) Principle

Equal quantities may be **added to (or subtracted from) both sides** of an equation without changing the solution or equality.

Solve for x.

a) $x - 15 = -5$

b) $14 = x - 3$

c) $1.5 + 0.2 = 0.3 + x + 0.2$

The Multiplication Principle

Both sides of an equation may be **multiplied** by the **same quantity (**except zero).

Both sides of an equation may be **divided** by the **same quantity** (except zero).

Note: When the coefficient of x is a fraction, we can get x alone by multiplying by the reciprocal of the coefficient.

Solve for x.

a) $2x = 20$

b) $\frac{x}{3} = 10$

c) $-3x = 48$

d) $\frac{5}{2}x = 15$

To solve linear equations of the form $ax + b = c$:

- Use the addition principle to obtain an equation that has the term containing the variable on one side and the constants on the other side.
- Use the division or multiplication principle to solve for the variable.

Solve.

a) $3x + 2 = 17$

b) $5x + 3 = -27$

c) $9 - 4x = 13$

d) $16 = 12 - 5x$

To solve equations with a variable on both sides of the equation:

- Combine like terms on each side of the equation.
- Use the addition principle to obtain an equation that has the terms containing the variable on one side of the equals sign and the constants (numbers) on the other side.
- Use the multiplication principle to solve for the variable.

Solve.

a) $5x + 3 = 3x - 17$

b) $2x - 5 = 5x + 7$

c) $0.6y + 0.8 = 0.1 - 0.1y$

To solve linear equations with parentheses:

- Use the distributive property to remove the parentheses
- Proceed as you did for the previous examples.
  - Combine like terms on each side of the equation.
  - Use the addition principle to obtain an equation that has the terms containing the variable on one side of the equals sign and the constants (numbers) on the other side.
  - Use the multiplication principle to solve for the variable.

Solve.

a) $4(x+1)-3(x-3)=2$

b) $2(3x-5)+2=4x-3(8+2x)$

c) $10+3(1+2x)=5[2x-(1-x)]$

1.6 Solve problems using inequalities

We often speak of one value as being greater than or less than another value. These relationships are called inequalities.

“6 is less than 12.”

“12 is greater than 6.”

These statements can also be written using mathematical symbols.

> represents the words “is greater than”

< represents the words “is less than”

Inequality	Symbol	Visual	Description
Greater than	>		All points to the right of the open circle are included in the solution set.
Greater than or equal to	≥		The circle and all points to the right of the circle are included in the solution set.
Less than	<		All points to the left of the open circle are included in the solution set.
Less than or equal to	≤		The circle and all points to the left of the circle are included in the solution set.

Inequalities may be solved in the same way we solve equations. However, if you multiply or divide by a negative number, the direction of the inequality must be reversed.

Note: To avoid any confusion, it is recommended that you move your variables to the left of the inequality when solving.

Solve the inequalities.

a) $3x + 7 \geq 13$

b) $5x - 2 > 18$

c) $3x - 8 > 7x + 4$

d) $2(3x - 2\,) \geq 3(1 - x) + 2$

Determine whether the inequalities are true or false when $A = 1, B = -2, \text{ and } C = 0.$

a) $A \geq B$

b) $-A < C$

c) $2A > -\text{B}$

d) $2A + B \leq 0$

1.7 Solve problems requiring the transposition of equations

We can rearrange formulas using the same method that was previously used to solve equations. Our goal is to isolate a certain variable so we will have to "get rid of" everything else.

Example:

If $D = L - N$, solve for L.

To solve for L, we must isolate L. In other words, we need L = something. To get L by itself, we must add N to both sides of the equation.

$$D + N = L - N + N$$

$$D + N = L$$

Solve for the indicated variable.

a) $4p - 3t = 1$ Solve for p.

b) $PV = nT + R$ Solve for T.

c) $A = \frac{1}{2}bh$ Solve for h.

d) $P = 2L + 2W$ Solve for W.

e) $F = \frac{9}{5}C + 32$ Solve for C.